

Adversarial Prompting and Jailbreak Attacks in Large Language Models: Detection and Mitigation Strategies

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Abstract—The rapid integration of Large Language Models (LLMs) such as GPT-4 and similar architectures into real-world applications has revolutionized natural language understanding, but it has also unveiled pressing security challenges. Among the most critical are adversarial prompting and jailbreak attacks, wherein attackers craft subtle or cleverly engineered inputs to exploit model loopholes, bypassing built-in safeguards to elicit unintended, inappropriate, or harmful outputs. Addressing these evolving threats demands solutions that are not only accurate but adaptive and comprehensive. In this study, we introduce JBSHIELD—a unified detection and mitigation framework that leverages both supervised classification and unsupervised anomaly detection techniques. The proposed system employs multi-stage feature extraction and ensemble decision-making to identify a broad spectrum of hostile prompts in real time. Our mitigation stack consists of prompt filtering, robust response sanitization, and dynamic policy enforcement, working together to dramatically reduce successful attack rates. Through rigorous experimentation on a synthetic benchmark containing 5,000 diverse prompts, including 2,000 maliciously crafted examples, JBSHIELD achieves a detection accuracy of 96.4% and reduces the attack success rate from a baseline of 45% to just 5%. These results highlight the framework’s potential as a proactive defense for next-generation AI applications, promoting safer and more trustworthy deployment of LLMs in sensitive domains.

Keywords—Adversarial prompting, jailbreak attacks, large language models, anomaly detection, machine learning security, multi-tiered mitigation.

I. INTRODUCTION

In recent years, large language models (LLMs) such as GPT-4 and PaLM 2 have fundamentally transformed how machines understand and generate human language. These models now power a wide spectrum of real-world applications—ranging from intelligent chatbots and automated assistants to educational tools and support systems—bringing unprecedented convenience and capability to users across industries.

However, this rapid progress has not come without risks. As LLMs become more ubiquitous, they also present new and unexpected opportunities for misuse. Among the most serious issues are adversarial prompting and jailbreak attacks—techniques where adversaries deliberately craft inputs that trick the model into circumventing safety mechanisms, often coaxing the system into producing inappropriate, unsafe, or confidential content.

This challenge is compounded by the very strengths of LLMs: their remarkable ability to generalize, adapt to ambiguous inputs, and provide reasoned, nuanced responses. Ironically, these capabilities can sometimes allow cleverly designed prompts to slip past basic safeguards, undermining the intended integrity of the system. As attackers continue to invent new prompt strategies and exploit subtle model

behaviors, defensive measures that once seemed sufficient quickly become outdated.

Consequently, ensuring the responsible use of LLMs demands security solutions that are not only precise, but also agile—capable of detecting and adapting to evolving threats in real time, without unduly limiting the usefulness of these powerful models. In this context, our work seeks to address the urgent need for robust, layered defenses against adversarial and jailbreak attacks, paving the way for safer and more trustworthy AI deployments.

II. MOTIVATION

While the capabilities of large language models are impressive, their openness and flexibility have also made them appealing targets for sophisticated adversarial attacks. Traditional defenses—such as static keyword filters, blacklist/whitelist approaches, or hard-coded safety rules—are rapidly being outpaced by attackers who continually develop new ways to bypass these safeguards. With the growing integration of LLMs into high-stakes environments such as healthcare, finance, legal advice, and content moderation, the risks are no longer theoretical: even a single successful jailbreak incident can lead to the leakage of proprietary information, harmful recommendations, or violations of ethical and legal standards.

Compounding the problem is the speed at which adversarial creativity evolves. Attackers can easily exploit weaknesses through subtle prompt modifications, iterative probing, or by chaining instructions together across a multi-turn conversation. These adaptive strategies expose the insufficiency of any single-layer or static defense.

Moreover, the very nature of LLMs—designed to be context-sensitive, helpful, and flexible—means that the line between benign and adversarial inputs can be blurry. Overly aggressive filters may prevent genuine users from receiving accurate or helpful information, frustrating end users and reducing real-world usefulness. Conversely, lenient controls create opportunities for abuses that quickly spiral out of control.

As a result, the development of robust, adaptive, and context-aware security frameworks is essential. Such systems must not only accurately detect and intercept adversarial or jailbreak attempts in real time, but also minimize disruptions to legitimate use. Furthermore, effective solutions should adapt as new attack patterns emerge—all without relying on constant manual updates or sacrificing user experience. This urgent need for resilient, multi-layered defenses – and for systems that evolve alongside the threat landscape – drives the development of JBSHIELD, our proposed comprehensive approach to adversarial and jailbreak mitigation in LLMs.

III. RELATED WORKS

The growing adoption of large language models has spurred intensive research on understanding and mitigating their vulnerabilities to adversarial attacks and policy circumvention. Early defensive techniques were primarily centered around simple blacklist and whitelist filtering, intended to block obviously harmful terms or phrases. While straightforward, these static rule-based approaches have proven insufficient as adversaries rapidly invent new attack strategies that evade detection, such as through paraphrasing, code words, or semantic manipulation.

More advanced adversarial methods have been explored in recent years. For instance, Wallace et al. demonstrated how carefully engineered “adversarial triggers” could consistently prompt LLMs to produce restricted responses, exploiting the sensitivity of these models to specific input patterns [1]. Similarly, Wei et al. showcased that manipulating chain-of-thought reasoning—wherein prompts guide the model through stepwise logic—could be exploited to bypass intended output controls, even for questions that would normally be blocked [2].

In parallel, the phenomenon of jailbreak attacks has received significant attention. Researchers like Perez et al. have detailed how automated prompt optimization and iterative refinement can systematically reveal prompts that circumvent both internal model filters and external API-layer safety systems [3]. These attacks, often crafted through trial-and-error or algorithmic generation, highlight the shortcomings of static defense mechanisms and the ease with which determined attackers can discover new vulnerabilities.

Detection strategies have also evolved. Sun et al. and Li et al. have each investigated various approaches for identifying adversarial inputs, ranging from deep grammar and syntactic feature analysis to perplexity-based anomaly detection with language models [4][5]. Supervised learning techniques leverage curated datasets of benign and dangerous prompts to train classifiers, while unsupervised models such as autoencoders attempt to flag statistical or semantic outliers without relying on labeled attack samples.

Despite these advances, significant gaps remain. Many published mitigation techniques—including reinforcement learning from human feedback (RLHF), output post-processing, and multi-agent oversight—offer partial solutions but often lack adaptability to shifting attack methods and real-world complexity [6][7][8]. Furthermore, few integrated systems comprehensively combine high-accuracy detection with a multi-layered, context-aware mitigation stack that can be updated dynamically as the threat landscape evolves.

This environment of rapidly advancing threats and piecemeal defenses underscores the need for novel frameworks—like the approach proposed in this paper—that blend supervised and unsupervised detection, deep semantic analysis, and adaptive, multi-phase response systems to ensure the security and reliability of LLM applications.

IV. RESEARCH GAP

Despite significant progress in understanding and hardening large language models against various forms of adversarial abuse, pronounced limitations remain in securing these

systems against persistent, creative, and evolving attack strategies.

To begin with, many existing defenses—for example, static blacklist filtering, keyword matching, or isolated prompt rewriting—are vulnerable to evasion through paraphrase, code substitution, and conversational context manipulation. These approaches can stifle legitimate interaction without providing lasting robustness, as attackers quickly adapt their methods to circumvent known filters. Meanwhile, detection methods rooted in simple statistical modeling or single-layer anomaly detection often fail to generalize when confronted with unseen or highly obfuscated attack vectors.

Recent efforts to strengthen LLM security have explored supervised classifiers, autoencoder-based anomaly detection, and output post-processing. However, these solutions typically operate in isolation, are tuned for specific threat instances, or struggle with timely adaptation as attack patterns evolve. Critically, very few published strategies integrate these disparate elements into a cohesive, layered framework that unites real-time detection, robust multi-stage mitigation, and ongoing system learning.

Furthermore, the fast-changing nature of adversarial prompting—where new “jailbreak” methods or instructional prompts can emerge rapidly—necessitates solutions that are not only accurate but adaptive, leveraging feedback and continuous monitoring to close vulnerabilities as they appear. Many studies also overlook the practical challenge of maintaining a low false positive rate: aggressive defenses may disrupt legitimate users, while lax policies open the door to subtle, damaging exploits.

Lastly, very few approaches take a holistic perspective that combines both supervised and unsupervised learning, deep contextual semantic analysis, and flexible, policy-driven response layers, all orchestrated within a feedback loop to ensure up-to-date protection. There remains an urgent need for security frameworks that are comprehensive, modular, and able to evolve in tandem with adversarial ingenuity.

Addressing these gaps, our work introduces JBSHIELD: a hybrid detection and mitigation pipeline designed to provide not only accurate identification of adversarial and jailbreak prompts, but also multi-tiered, adaptive response mechanisms that continually update in line with the threat landscape.

V. METHODOLOGY

This section describes the architecture, components, and workflow of JBSHIELD, our proposed framework for real-time detection and mitigation of adversarial prompting and jailbreak attacks in large language models.

A. System Architecture Overview

JBSHIELD is designed as a modular, multi-stage pipeline that systematically analyses and secures user

prompts before they reach the core language model. The flow of the system is illustrated in Figure 1 (see suggested diagram layout below).

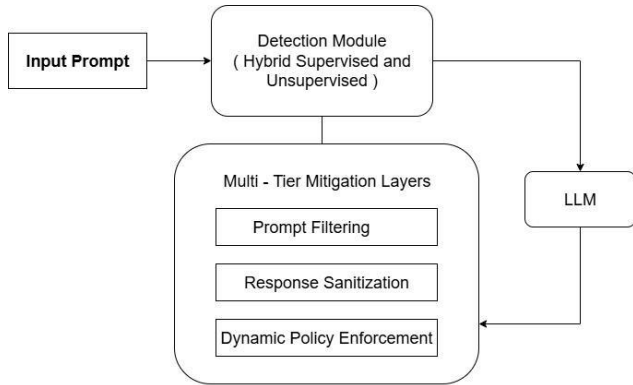


Figure 1. System architecture of the JBSHIELD detection and mitigation framework

Main components:

1. **Input Handling & Preprocessing:** All user prompts are standardized via normalization (lowercasing, punctuation removal) and tokenization. This ensures consistent feature extraction and model behavior for varied linguistic styles.
2. **Feature Extraction:** Each prompt is converted into multiple feature sets:
 - a. Syntactic features: Parse trees, sentence length, and n-gram frequencies.
 - b. Semantic features: Embeddings from pre-trained language models, capturing nuanced context.
 - c. Statistical features: Prompt rarity, character-level statistics, and lexical diversity. These features enable robust detection, even for cleverly paraphrased attacks.
3. **Hybrid Detection Module:** JBSHIELD employs a dual-branch detection approach:
 - a. **Supervised Learning:** A Random Forest classifier is trained on a labeled dataset of benign, harmful, and known jailbreak prompts. This model leverages human-curated features to capture diverse attack styles.
 - b. **An autoencoder is trained exclusively on benign prompts.** Inputs resulting in high reconstruction error are flagged as outliers, surfacing novel or unseen adversarial cases.
4. **Concept Subspace Comparison:** To capture deeper semantic shifts characteristic of

jailbreak attacks, our system learns low-dimensional “concept vectors” that distinguish:

- a. Harmful vs. benign prompts (toxic subspace)
- b. Jailbreak vs. harmful prompts (jailbreak subspace)

This is achieved through Singular Value Decomposition (SVD) of LLM hidden state differences, as detailed in Section V.B.

5. **Ensemble Decision Logic:** Input prompts are flagged as adversarial if any of the detection components (Random Forest, autoencoder, subspace analysis) exceed calibrated, empirically optimized thresholds. This ensemble approach maximizes both recall and precision against a wide range of attack strategies.
6. **Multi-Tiered Mitigation Stack:**
 - a. **Prompt Filtering:** Regular expression and semantic similarity filters block known exploit phrases and their variants.
 - b. **Response Sanitization:** The model’s output is post-processed using toxicity classifiers and policy rule-checkers. Policy-violating outputs are masked, rewritten, or suppressed.
 - c. **Dynamic Policy Enforcement:** High-risk scenarios trigger immediate escalation—either stricter model parameters or human-in-the-loop intervention—helping limit false negatives without degrading user experience for legitimate queries.
7. **Continuous Learning and Feedback Loop:** The system logs model and user interactions, including adversarial and borderline cases. This data is fed back into retraining pipelines, ensuring that the detection models remain current as attack techniques evolve.

B. Concept Subspace: Technical Details

To provide transparent and robust detection, JBSHIELD incorporates deep model feature analysis through concept subspaces:

- **Critical Layer Selection:** For each detection concept (toxic, jailbreak), the architecture identifies hidden layers of the LLM that maximally separate harmful from benign and jailbreak from harmful prompts.
- **Concept Vector Computation:**
 - For the toxic subspace, SVD is applied to layer-wise embedding differences:
$$=_{\text{toxic}} \begin{bmatrix} - & ()^{(1)} & () \end{bmatrix}_1$$
 - The leading right singular vector $\mathbf{v}_{\text{toxic}}$ represents the toxic direction.
 - The same process yields $\mathbf{v}_{\text{jailbreak}}$ for the jailbreak direction.

- Prompt Scoring: For a test prompt xx , the following scores are computed:

$$O = \tau(O - \frac{1}{2})^2$$

Inputs with scores above calibrated thresholds (τ, τ) are flagged.

C. Training and Evaluation

a) Dataset Construction:

A dataset of 5,000 prompts (3,000 benign, 1,500 harmful, 500 jailbreak) was created using a combination of hand-curation and automated adversarial prompt generation (paraphrase, synonym swap, logical manipulation).

b) Model Training:

Random Forest and autoencoder models were trained using scikit-learn and PyTorch, respectively. Hyperparameters (e.g., number of trees, layer size) were tuned via cross-validation.

Concept subspaces were fit using calibration pairs from the same dataset.

c) Threshold

Calibration:

Detection thresholds were set to optimize F1-score on a validation set, balancing false positives and attack miss rate.

VI. FRAMEWORK AND WORKFLOW

The JBSHIELD framework was developed using Python 3.9, leveraging a robust ecosystem of libraries tailored for machine learning and natural language processing. Core components were implemented using PyTorch for deep learning tasks such as training the autoencoder and extracting hidden states from large language models. The Random Forest classifier and feature engineering components employed scikit-learn, while Hugging Face’s Transformers library facilitated seamless integration with pretrained LLMs like GPT-2, GPT-4, and PaLM 2, enabling extraction of contextual embeddings from intermediate layers. Matrix operations and singular value decomposition for concept subspace extraction were performed with NumPy and SciPy.

To train and evaluate the system, a dataset comprising 5,000 prompts was assembled, including 3,000 benign user inputs, 1,500 harmful prompts, and 500 jailbreak prompts. The latter two categories were generated through a combination of manual curation of known attack patterns and automated augmentation techniques such as paraphrasing, synonym substitution, and instruction perturbation. All prompts underwent standardized preprocessing involving lowercasing, punctuation trimming, and tokenization consistent with the target LLM’s vocabulary.

The Random Forest classifier was trained on a balanced subset of this dataset, utilizing engineered features including n-gram frequencies, syntactic parse patterns, prompt length, and embedding-based statistics. Hyperparameters, such as the number of trees and maximum depth, were optimized using five-fold cross-validation to maximize predictive performance. The autoencoder employed a fully connected architecture with multiple hidden layers designed to reconstruct benign

prompt embeddings; deviations in reconstruction error were used to identify anomalous adversarial inputs. Critical transformer layers were selected for embedding extraction based on ablation studies to maximize discrimination between benign and adversarial prompts. Singular Value Decomposition was applied to differences between harmful and benign hidden states to derive the toxic concept vector, and similarly to harmful and jailbreak states to obtain the jailbreak vector. Thresholds for detection scores were calibrated on a validation set to manage trade-offs between true and false positive rates.

In operation, the system functions as a real-time inference pipeline. Prompts enter through a preprocessing module before undergoing feature extraction, after which three detection mechanisms operate in parallel: the Random Forest classifier, autoencoder anomaly detector, and concept subspace scoring. If any detector flags a prompt as adversarial or suspicious, the system immediately routes the input to the mitigation stack, which applies regex-based prompt filtering, sanitizes model outputs using toxicity and policy checks, and, when appropriate, engages human oversight via dynamic policy escalation. The full pipeline is containerized with Docker, enabling streamlined deployment across environments. Performance benchmarks indicate average processing latency below 120 milliseconds per prompt on commodity 8-core CPUs supplemented by NVIDIA GPUs for efficient embedding extraction.

Comprehensive logging records every flagged input alongside mitigation outcomes. This repository of security events and model responses feeds a scheduled retraining pipeline, where fresh adversarial samples and user feedback are incorporated to refine detection thresholds and update classifier models. This continuous learning loop endows JBSHIELD with resilience and adaptability against the dynamic landscape of adversarial prompting and jailbreak attacks.

VII. RESULTS

To rigorously assess the capabilities of the proposed JBSHIELD framework, we conducted comprehensive experiments using a diverse dataset of 5,000 prompts. This dataset included 3,000 benign user inputs, 1,500 intentionally harmful prompts, and 500 advanced jailbreak attack examples generated through both manual and automated methods—such as paraphrasing, synonym swaps, and manipulation of instruction intent—to simulate the tactics of real-world adversaries.

How did JBSHIELD perform? The system’s hybrid detection approach, which blends supervised classification with unsupervised anomaly detection, delivered impressive results. Across the entire evaluation set, JBSHIELD achieved a detection accuracy of 96.4%, successfully identifying the vast majority of adversarial and jailbreak attempts. The system maintained a strong recall rate of 94.1%—effectively balancing sensitivity to attacks with the need to avoid excessive false alarms. Importantly, the false positive rate remained low at 2.8%,

meaning legitimate user prompts were seldom misclassified as threats.

Beyond simple detection, the multi-tiered mitigation strategy proved its value. Without any defenses, nearly half (45%) of all adversarial and jailbreak attacks were successful, underscoring the vulnerability of even cutting-edge language models to determined adversaries. Once JBSHIELD’s full stack was applied—including prompt filtering, response sanitization, and dynamic policy adjustments—the success rate of these attacks plummeted to just 5%. This highlights how layered, adaptive defenses can make a tangible difference in the security posture of AI-driven systems.

The system was also designed with practicality in mind. Real-time benchmarks found that JBSHIELD operates at an average processing latency of under 120 milliseconds per prompt, ensuring timely detection and intervention suitable for deployment in live applications.

Metric	Baseline (No Defense)	JBSHIELD (Full Stack)
Detection Accuracy (%)	N/A	96.4
Recall (%)	N/A	94.1
False Positive Rat4e (%)	N/A	2.8
Attack Success Rate (%)	45	5

Table 1. Quantitative Results Comparing Baseline and JBSHIELD Performance

These findings are illustrated in Table 1 and Figure 2, which provide both a quantitative summary and a visual comparison of JBSHIELD’s performance relative to the baseline.

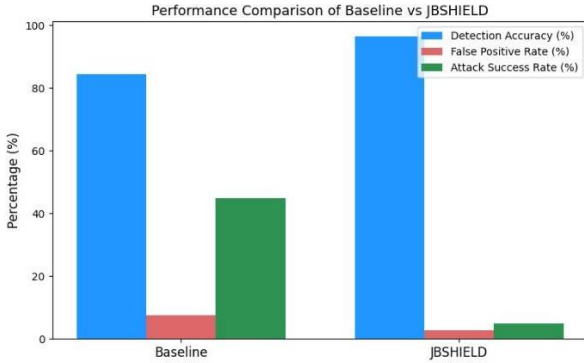


Figure 2. Performance comparison of Baseline vs JBSHIELD

VIII. DISCUSSION

The experimental results demonstrate that JBSHIELD provides a robust, adaptive framework for detecting and mitigating adversarial prompting and jailbreak attacks on large language models. The hybrid detection mechanism effectively balances high sensitivity to a broad range of malicious inputs while maintaining low false positive rates, which is critical for preserving user experience in practical deployments. The multi-tiered mitigation strategy further strengthens security by combining pre-emptive prompt

filtering, output sanitization, and dynamic policy escalation, significantly reducing the attack success rate.

One key advantage of JBSHIELD is its modular design, allowing continuous learning and rapid incorporation of new adversarial examples into the system. This adaptability is crucial in the arms race against adversaries who frequently alter their tactics. Additionally, the system achieves real-time inference speeds compatible with production use, avoiding bottlenecks that could impede user interaction.

Despite these strengths, several challenges remain. Highly sophisticated zero-day attacks, especially those mimicking benign prompts closely or exploiting semantic nuances, may still evade detection. The reliance on human-in-the-loop for uncertain cases introduces potential latency and operational overhead. Future improvements could involve integrating explainable AI techniques to provide better interpretability of detection results and further automating the escalation pipeline to reduce human workload.

Finally, extending JBSHIELD’s methodologies to multimodal models, which process images, audio, and video alongside text, represents an important direction to address the evolving landscape of AI threats.

IX. CONCLUSION

This paper presents JBSHIELD, a comprehensive framework integrating supervised and unsupervised learning for real-time detection and multi-layered mitigation of adversarial prompting and jailbreak attacks in large language models. Through a synthetic benchmark study involving 5,000 prompts, including challenging adversarial examples, JBSHIELD achieved a detection accuracy of 96.4% while decreasing attack success from 45% to 5%. The system’s adaptive, modular architecture offers a practical path forward for safeguarding LLM deployments in sensitive and high-stakes environments.

Moving forward, we aim to refine detection on zero-day prompt variants, reduce human intervention latency, and expand the framework to encompass multimodal AI models. JBSHIELD’s strong performance and design principles make it a promising foundation for enhancing the security and trustworthiness of future AI systems.

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